Power Electronics Report2

by Jin Lian

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Power Electronics Report

BOOST CONVERTER AND CONTROL DESIGN

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2 – Basic Converter Design

a) Choosing an inductor value for the boost SMPS operating under continuous conduction operation whilst neglecting the component nonidealities is fairly straightforward. The inductor value depends on the input voltage (Vin), allowable inductor current ripple (ΔI_L), duty cycle (δ) and switching frequency (f) via this relationship:

$L = \frac{\delta * Vin}{f * \Delta I_L}$. To determine the minimum inductance required we had to optimise the parameters in this relationship to get the minimum L. The duty cycle was fixed at 7/12 ($\delta = 1 - \frac{Vin}{Vout}$) from the given input (20V) and output voltage (48V). f was maximised to the maximum limit of 200KHz. ΔI_L was maximised as a maximum proportion of $I_{in} * (0.1 * I_{in})$. I_{in} was then maximised based on the minimum efficiency α allowed ($P_{out} = \alpha P_{in} => V_{out} * I_{out} = \alpha V_{in} * I_{in}$). The maximum Iout (0.6A) was specified hence α was minimised (α = 0.9) to maximise ΔI_L and thus minimise L. Based on these relationships and parameters Lmin ≈ 0.3645mH. $L_{min} = \frac{\delta * Vin}{f_{max} * \Delta I_{L_{max}}} = \frac{\frac{7}{12} * 20}{200000 * 0.1 * 1.6} \approx 0.365mH$

b) The peak current is important because it tell us the maximum instantaneous current flowing in the circuit. This is a necessary parameter to choose component max. ratings e.g. I_{DS} max. for the MOSFET and I_{FB} max. for the diode.

The RMS current is equivalent to the DC current that will produce the same average power dissipation in a resistive load. This parameter is essential to determine the max. power ratings of the components especially the MOSFET and diode.

i) $I_{L(PEAK_WORST)} = I_{IN} + \frac{\Delta I_L}{2} = 1.05 * I_{IN} = 1.05 * \frac{Vout * Iout}{\alpha_{min} * Vin} = 1.05 * \frac{48 * 0.6}{0.9 * 20} = 1.68A$

ii) $I_{L(RMS_WORST)} = \sqrt{I_{L(AVG_MAX)}^2 + \left(\frac{\Delta I_L}{2\sqrt{3}}\right)^2} = 1.6\sqrt{1^2 + \frac{0.05^2}{3}} \approx 1.60067A$

ci) Choosing the minimum output capacitance value:

$\Delta V_{C_CAP} = \frac{\delta * I_o}{f * C}$. $\Delta V_{C_ESR} = \Delta I_C * R_{ESR}$

Capacitor C is minimised by maximising ΔV_{C_CAP}, i.e. by taking the fundamental ripple voltage across the capacitor to be due to its capacitance rather than its effective series resistance (ESR). Minimising duty cycle (δ) and output current (I_o), and maximising the switching frequency (f) will also reduce C.

$C_{min} = \frac{\delta * I_o}{f * \Delta V_{C_CAP}} = \frac{\frac{7}{12} * 0.6}{200000 * 1} = 1.75\mu F.$

di) $I_{D\,max} = I_{L(PEAK)\,max} = 1.05 * I_{IN} = 1.68A$ i.e. the diode's repetitive peak forward current should exceed this value. However, during start-up there will most likely be an inrush of current especially if C is large and the load is also small (i.e. of high resistance), hence the input current and diode current will exceed their steady state value by a significant amount. Thus, the diode must be able to withstand a peak forward surge current of several times this value.

The diode should also be able to withstand a repetitive reverse bias voltage equal to the output voltage i.e. 48V. Again, if the system is operated in open loop and isn't sufficiently damped Vout may have a large percentage overshoot at start-up hence the diode needs to be able to withstand peak reverse bias voltages of more than 48V e.g. 60V.

For similar reasons to the diode mentioned above, the output capacitor must be able to withstand a potential difference of 48V across its plates and a peak potential difference of around 60V or more across its plates if the boost SMPS plant is to be operated in open loop mode. Capacitors with a voltage rating of 63V or more will be ideal.

3 – Simulation of Ideal Converter in Open-loop

Using ideal components given by OrCAD capture, we built an ideal open-loop boost converter as shown in the following graph.

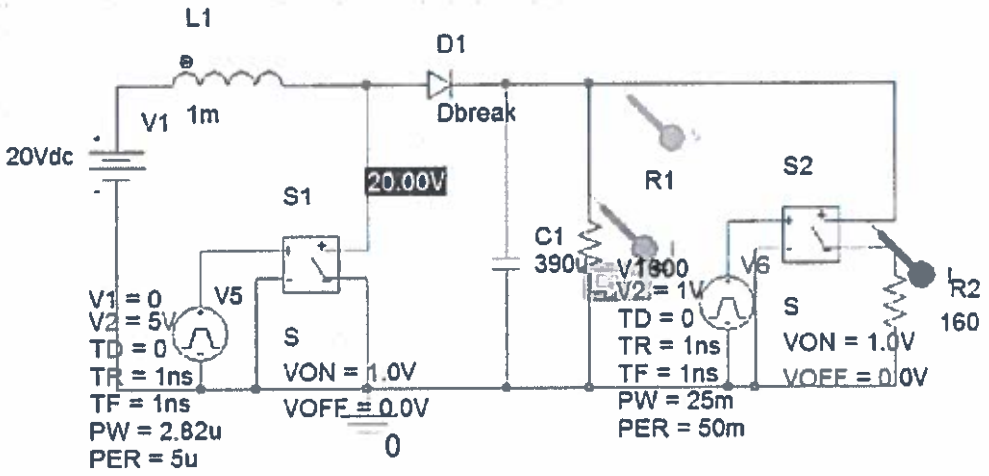


Figure 1: Open Loop Boost Converter with Ideal Components

With the circuit shown in figure 1, several simulations were run and important wave forms were saved for further analysis (see appendix), also some important values were recorded in order to analyse key parameters for comparison.

Table 1 (see appendix) shows the efficiency under different loads connected to output. As the table suggests, as Rload increases, Vout increases and efficiency increases as well. This is what should happen because when Rload increases, I out decreases, less current is drown from C1 (output capactor), therefore Vout is higher. Also, when Rload increases, I in decreases, which makes the only loss in the circuit (i.e. diode voltage drop) smaller, therefore efficiency is improved.

Note that we are still in continous conduction mode until R = 640Ohm, for R = 1280 Ohm. We are just in discontinous mode, when in discontinous mode, Vout will keep rising because for discontinous mode $\frac{V_o}{V_i} = \frac{1}{1-\frac{V_i \delta^2}{2fLI_i}}$, decrease Iin in will increase Vout. but efficiency decreases.

More presise measurements were made for R load = 80 (Max load) and steped load from 80 to 160.

From the table 2 (see appendix) (readings in table are directly obtained from cursor reading in pspice.) we could see with Vout overshoot is 10.18%, which is fits design specification. The ripple in Vout is 0.004V, which is negligible.

The rise time is measured from V = 0 to Vmax because our design is underdamped.

Oscillation frequency is $200kHz$, which is set by frequency of Vpulse, since vpulse has period of $5\mu s$, the frequency should be $\frac{1}{5\mu} = 200klz$, just as it is from pspice simulation.

When there is a step change in load from 100% to 50%, we could see voltage rises and current declines like for when starting-up the converter, both of them will oscillate to and converge to a steady state value for new load. There are overshoot/undershoot for them because of this step change which depends on transfer function.

From table 3, when we have very small Rload connected to the system, Vout drops and I in increases significantly (from 43.3V/1.24A to 38.8V/4.45A). Also, efficiency is seriously affected as well. However the overshoot and settling time become smaller because with smaller Rload (i.e. greater I out) could discharge capacitor quicker. Rise time doesn't change due to the increase in maximum inrush current and larger output current cancels out with each other.

4 – Real Component Choices

Inductor:

Ideally the larger the inductance chosen the better i.e. smaller current ripple and output capacitor voltage ripple for a given frequency, hence we can use a lower frequency and reduced the switching losses of the MOSFET or use a lower capacitance output capacitor. Unfortunately, large inductors are less ideal and come with significant series resistance which degrades the efficiency of our boost SMPS. We chose our inductor to be cheap, of high inductance and with low ESR. An inductance of >1mH of low ESR (<1Ω) was attractive, based on the lower limit (0.365mH) obtained from our calculations earlier.

Diode:

We required a diode as close to ideal as possible i.e. having low capacitance, low reverse recovery time (fast switching speeds), low forward voltage and with a suitably high or forward biased max current and reverse biased voltage rating. To achieve low forward biased voltage to increase efficiency we required a rectifying schottky diode, which has a lower forward bias voltage (approx. 0.4-0.6V) than standard pn silicon diodes (0.7-1V). Schottky diodes also have the advantage of much shorter reverse recovery times than standard pn silicon diodes, however they have a much higher reverse bias leakage current (approx. 10 times greater). We chose to use either a fast revovery rectifying pn diode or a schottky diode to ensure proper operation at a high switching frequency of 200KHz whilst reducing losses.

Capacitor:

The larger the capacitance chosen for the output capacitor the better. Larger capacitance results in lower output voltage ripple for a given switching frequency. Larger capacitors also tend to have lower ESR. Hence, choosing a large output capacitance is very beneficial. However, a large reduces the speed of response and start-up time of the system. It also results in a large input current in-rush at start up, particularly for smaller loads (lower resistance). This is a negative effect, which must be avoided. Consequently, we chose C to be sufficiently bigger than the min C (1.75μF) calculated earlier on in order to obtain the desired output voltage ripple of $< \pm 0.5V$. A separate current limiting feedback circuit can be used to limit this input current in-rush at start up (i.e. providing a soft start), hence making the choice of C independent of the maximum input current allowed.

Transistor

The main losses incurred by the switching transistor are due to switching losses (which result from its finite rise and fall times) and conduction losses (which result due to its finite drain-source on resistance, $R_{DS(ON)}$). Conduction losses are proportional to $R_{DS(ON)}$ and switching losses are proportional to frequency. We required a N-channel enhancement MOSFET capable of operating properly at a switching frequency of 200KHz. Next we chose a MOSFET with a low $R_{DS(ON)}$, of the same order of magnitude as resistances in the circuit i.e. <10mΩ.

MOSFET Gate Drive

In order to switch MOSFET faster to reduce power loss when switching, MOSFET Gate Drive should be used if the control signal couldn't provide large current. Longer turn on/off time means greater power loss and produces more heat. For our real component simulation, we use ideal voltage source to control MOSFET gate, which makes this implementation not needed. However when we start to build our design in real world, we may need this implementation.

Diode	Transistor	Capacitor	Inductor
1N5806	IRFB7546	390uF 0.027Ohm	1mH 0.419Ohm

5 – Simulation and Analysis with Real Components

Based on the selection from section 4, datasheets were found and models were updated to orCAD pspice simulation program.

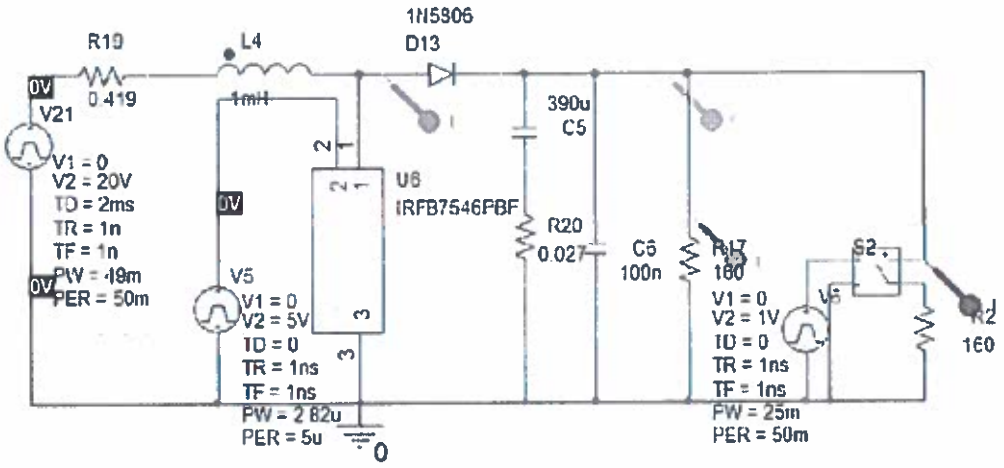


Figure 2: Open Loop Boost Converter with Real Components

This is now an open-loop simulation for real components with parasitic resistances added and real component model updated.

As stated in section 3, simulations were run and important wave forms were saved for further analysis (see appendix), important values were recorded in order to analyse key parameters for comparison.

Table 5 shows the efficiency given by different output loads. As Rload increases, Vout and efficiency increase. It behaves the same way as ideal model values . The larger the Rload the smaller the current flow inside converter, which will make the power losses from transistors, diodes and resistors smaller. However, when Rload = 1280 Ohm, the device is still in continous mode. This may because the introduction of parasitic resistance makes current ripple smaller.

It is quite difficult to exam the efficiency when the device is driven into discontinous mode because of simulation time. It takes nearly 150ms for R = 1280 Ohm to settle to steady state. If larger R are simulated, longer time it needs and simulation data will occupy more space which makes simulation impossible. However, we could see when driving into discontinous mode, the efficiency will decrease and V out will keep increasing.

Again, more presise measurements were made for R load = 80 and a steped load from 80 to 160

As shown in table 4 and graph in appendix (appendix number), input current stops at 0A but going further down to small negative figure. This may rise because diode is not ideal, therefore it will have some leakage current going through it when transistor is turned off.

Percentage overshoot is 27.77%, which is higher than ideal model, and maximum in-rush current is much higher. It takes longer to rise and settle as well.

The argument for step change in load holds the same for real components compared to ideal circuit.

There is quite a large spike current flowing through diode (graph in appendix) (appendix number). The negative part is recovery current and the positive part is due to the loss of parasitic inductance in circuit. Recovery current is formed because excess charge in n region needs to be removed when diode turning into reverse bias.

The oscillation and huge spikes in Vout and Iout are introduced because the transients when turning off diode.

Table 6 shows when we have very small Rload connected to the real component system. It follows the discussion for ideal system.

6 – Synchronous Rectification

This technique involves replacing the diode in our boost converter with an active switch (i.e. another NMOS) to improve efficiency by removing the 0.7V voltage drop associated with a diode when conducting. A MOSFET has very low drain-source resistance ($R_{DS(on)}$ can be <0.01Ω) when conducting and very high resistance when off, hence it approximates an ideal switch. The problem associated with replacing the diode with a MOSFET is that the source terminal of the NMOS is connected to the drain of the switching NMOS i.e. this terminal is floating and not connected to a fixed voltage. In Pspice this problem can be easily avoided by connecting a pulse source to its source and drain terminals. In reality this problem isn't easily solved. The NMOS's are connected in a similar configuration to a half bridge driver circuit and hence the solution to this problem can be obtain from the half bridge driver circuit.

Various Half-Bridge Drive Circuit topologies:

See appendix.

As shown in the various diagrams in the appendix, there are numerous ways to implement a driver for boost converter to achieve synchronous rectification based on the half-bridge driver circuits. A

level shifter is required to switch the transistors in anti-phase ensuring both transistors aren't on simultaneously. If this occurred the output capacitor of the converter will be temporarily shorted to ground resulting in a significantly reduced overall efficiency. The transformer coupled model involves switching the NMOS via a transformer with two separate taps for each NMOS. The top NMOS is linked via additive polarity and bottom NMOS via subtractive polarity so they switch in anti-phase. The discrete BJT driver uses an push-pull stage to drive the NMOSs and an inverting amplifier to provide an anti-phase switching. When the top NMOS switches on its source voltage will be Vdd and Vg > Vs + Vth (Vth > 0) to switch it on. Hence, both the discrete BJT and IC driver require a higher power positive power supply voltage to switch the top NMOS. To overcome this problem a boostrap capacitor is used to switch the top NMOS. When the bottom NMOS is on its drain voltage will be equal to -Vss and V_{CB} will be 12V. When the top NMOS switches on its source voltage rises from -Vss, this rise is reflected to V+ of the top driver causing V+ to become Vdd + 12V temporarily. This provides the extra gate voltage necessary to satisfy Vg > Vs + Vth, hence switching the top NMOS on. The zener diode provides a path for the excess charge on CB to the 12V+Vss rail, hence returning V_{CB} to its initial value.

6.1 – Simulation of Synchronous Rectification

Replacing diode with MOSFET and controlled pulse voltage in last simulation gives us this circuit.

The pulse voltage are set carefully in order to prevent turning both MOSFETs on at the same time, because if both MOSFETs are on, we are effectively short output capacitance which will cause drop in Vout and efficiency.

Tested is done with maximum load and values are listed in table 8.

General speaking, we increase the efficiency by 1.18% and output voltage by 0.5V replacing diode with synchronous rectifier, it works well in continuous mode.

From graph in appendix (appendix number), we could further increase efficiency by shortening the off-off time (when both MOSFETs are off). The diode current spike for synchronous rectifier is larger than that for diode.

Of the same reason given in real component simulation, we couldn't test discontinuous mode for synchronous rectifier. From knowledge learnt, we can see how it will work in discontinuous mode.

When we drive the device into discontinuous mode, current through inductor decrease to zero, which will set the source voltage of diode-MOSFET to V_in which is 20V. Since V_in<V_out, and diode-MOSFET is still on, there will be a huge reverse current drown from capacitor through inductor and cause Vout and efficiency drop significantly. Therefore, when inductor current just fall into zero, we should turn off diode-MOSFET to prevent that happen.

Compare the efficiency with the efficiency obtained from real component simulation (table 7 and table 5), we could conclude that the efficiency will increase if we use synchronous rectification to replace diode in continuous mode.

However, if we replace diode with synchronous rectifier for further design, the time taken for each simulation will be quite long. And since it doesn't really increase efficiency much. We will still use

diode for further design. In real circuit we will be building, synchronous rectifier will be used to increase efficiency.

7 – Closed-Loop System

7.1 – Static and Dynamic Converter Model

Equations describing boost SMPS in continuous conduction mode whilst including the effects of a non-ideal diode voltage drop:

$$\Delta I_L(ON) = \frac{V_I * t_{ON}}{L} = \frac{V_I * \delta}{f * L}$$

$$\Delta I_L(DIODE) = \frac{(V_I - V_O - V_D) * t_{off}}{L} = \frac{(V_I - V_O - V_D) * (1 - \delta)}{f * L}$$

In steady state $\Delta I_L(ON) + \Delta I_L(DIODE) = 0$ i.e.

$$(V_I - V_O - V_D) * (1 - \delta) = V_I * \delta$$

$$V_I = V_O * (1 - \delta) + V_D$$

$$\frac{V_O}{(V_I - V_D)} = \frac{1}{1 - \delta}$$

Hence, to get the same output voltage for a given duty cycle we need to apply a larger input voltage $V_I + V_D$.

Derivation of state space averaging model of the boost SMPS around the set duty cycle at full rated load:

The switch, inductor and capacitor are assumed to be ideal.

See appendix.

Small signal open loop transfer function model H(s) of boost SMPS around a duty cycle of 7/12:

$$H(s) = \frac{-3692s + 5.128 * 10^{20}}{s^2 + 32.05s + 4.452 * 10^5}$$

H(s) has 2 poles at $-16 \pm j667$ and a zero at 13889. Since the poles of H(s) are all in the left half plan hence the open loop system is stable. H(s) is type 0 since it has no free integrators i.e. 1/s. Hence, the open loop system has a non-zero steady state error $e_{ss}(t)$ to a unit step, $e_{ss}(t) = H(0) \approx 115.2$. An integrator must be introduced as part of the compensator C(s) into the closed loop transfer function $\frac{K*C(s)*H(s)}{1+K*C(s)*H(s)}$ to reduce the steady state error of the closed loop system to 0. The damping factor $\zeta = \frac{32.05}{2*\omega_n} = \frac{32.05}{2*\sqrt{4.452*10^5}} \approx 0.02402$. Since $0 \leq \zeta < 1$, the open loop system is

underdamped. The settling time $t_s = \frac{4}{\zeta \cdot \omega_n} = \frac{4}{0.01826 \cdot \sqrt{4.452 \cdot 10^3}} \approx 0.249s$.

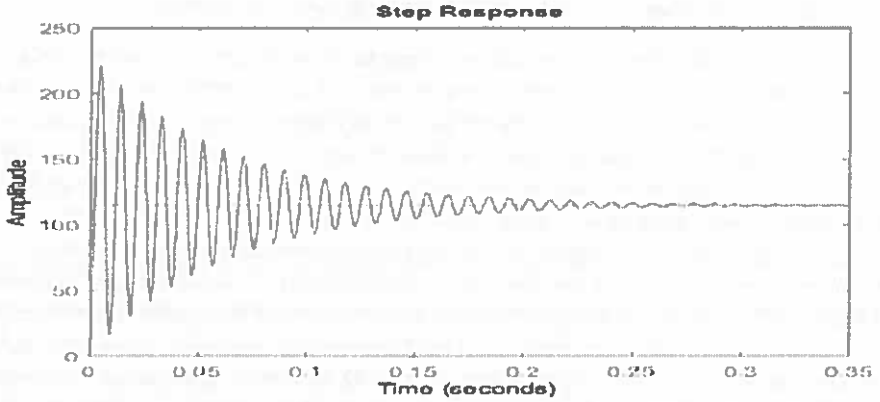


Figure 3: Step response of H(S)

See appendix for root locus of H(s).
The results in the previous section are confirmed from the step response of the open loop system.

7.2 - Compensator Transfer Function Design

In reality on the Pspice simulation software the compensator specified by Matlab based on the state space averaging model was a good starting point but not the best for our system. The matlab compensator model didn't perform as expected for several reasons. Firstly, the state space averaging model is a linearization of a non-linear model of the boost SMPS which has two independent matrix models i.e. one each for the ON and OFF states of the SMPS. The boost SMPS is a non-linear time invariant (LTI) system because the switching of the SMPS results in time-varying circuit equations. This state space averaging model is limited to operating at a fixed duty cycle > 0 and only when the perturbation applied is small. For a different set duty cycle the elements of the system matrix will be different and for large perturbations the output voltage response will be inaccurate. This model also provides no information on the switching frequency ripple dynamics. In addition, the state space averaging model we derived ignored non-ideal effects such as the series resistance of the inductor and capacitor. However, the simulations we ran on Pspice used non-ideal component representations and took into account the switching frequency ripple dynamics of the boost SMPS.

The main aims of introducing a compensator under closed loop operation of our boost SMPS is to enhance, provide and improve certain qualities of the system. Our system is required to have a steady-state output voltage equivalent to the desired value (48V) for varying loads. A soft start is necessary to limit the input current inrush to the required safety bounds. Similarly, the output voltage must not exceed a certain limit which may cause failure of certain appliances connected to the SMPS on start-up. Hence sufficient damping of the converter is essential to minimise percentage overshoot to tolerable limits.

To improve steady state error, we require a free integrator i.e. a phase lag compensator or integrator. To limit overshoot, sufficient damping is required. This can be provided by increasing phase margin (i.e. using a lead compensator or differentiator). A lag compensator can also improve phase margin by reducing the gain at high frequencies when the phase starts to approach -180°. Increasing rise time (i.e. speed of response) requires less damping whilst to reduce overshoot

requires more damping. There is an obvious conflict between these two properties, hence the converter's percentage overshoot is chosen to be < 10%. Providing a soft start by limiting input current in rush could be achieved by using limiting the input current at start-up.

Soft starting the converter: The inductor current is equal to the input current. A current inrush occurs at start up when the rate of increase in inductor current surpasses the rate of decrease in inductor current. This generally occurs when the product of R and C is large i.e. for large output capacitors and small loads. On start-up Vout is 0 hence the rate of decrease in inductor current is low compared with its rate of increase, so the inductor current always rises initially. When R*C is large the capacitor takes more time to charge hence Vout remains low for longer allowing the inductor current to continue ramping up. Eventually Vout becomes large enough and the net rise in inductor current becomes negative and the inductor current starts to ramp downwards towards steady state. If R*C is low the steady state inductor current value will be close to its maximum value since the rate of capacitor discharge makes Vout tend to drop and hence more current is required to result in a net rise in current of 0. To limit this current inrush for large R*C, a current sense resistor (R_{tense}) can be included with a feedback mechanism which consists of a comparator which senses the voltage across R_{sense} and starts to reduce the duty cycle if the input current becomes too large. R_{sense} can be very small (e.g. 0.01Ω) hence it should have little impact on the converter's efficiency. By reducing the duty cycle for large input currents, the MOSFET is switched on less and the diode conducts more. This results in less time for the output capacitor to discharge via the load i.e. keeping Vout more constant and making the rate of inductor current decrease larger and occur sooner.

Closed loop control – Compensator design using Matlab

PI (Proportional and Integral) controller results:

PI controller transfer function: $K_p + \frac{K_I}{s} = \frac{0.218}{s}$

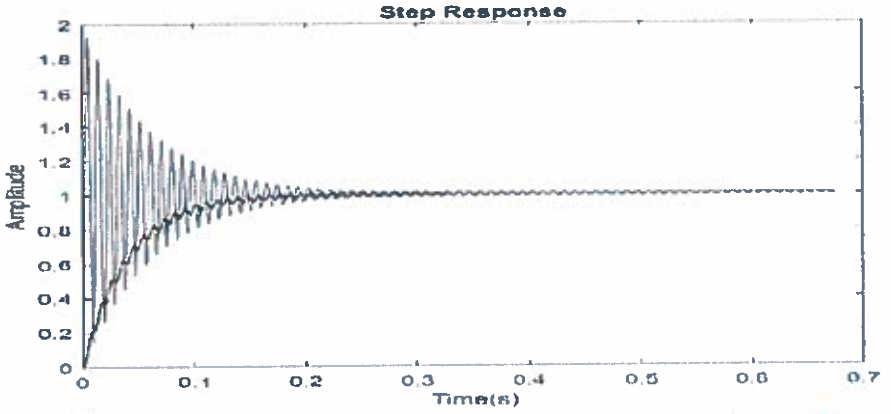


Figure 4: Red line is the open loop step response normalised by its steady state value. Black line is the closed loop step response using PI controller.

Using the Matlab pidtool it was possible to obtain a decent transient response with a PI (Proportional and Integral) controller in which the proportional gain (K_p) was set to 0 and the integral gain (K_I) was set to 0.218. The introduction of the compensator C(s) = 1/s (and with gain K = 0.218)

created a closed loop system with 3 poles relatively close together (i.e. 3 dominant poles) which provided a very similar step response to the open loop system. The introduced compensator and gain to the feedback loop removed the oscillations (i.e. the system became effectively overdamped $\zeta > 1$), and set the steady state error to 0. However, the settling time only slightly decreased and the speed of response was a bit faster. To obtain a much faster speed of response (and shorter settling time) whilst still maintaining an overdamped system a PID controller (Proportional, Integral and Derivative) controller was required.

PID (Proportional, Integral and Derivative) controller results:

PI controller transfer function: $K_P + \frac{K_I}{s} + K_D s = 0.00625 + \frac{1.49}{s} + 6.56 * 10^{-6} s$

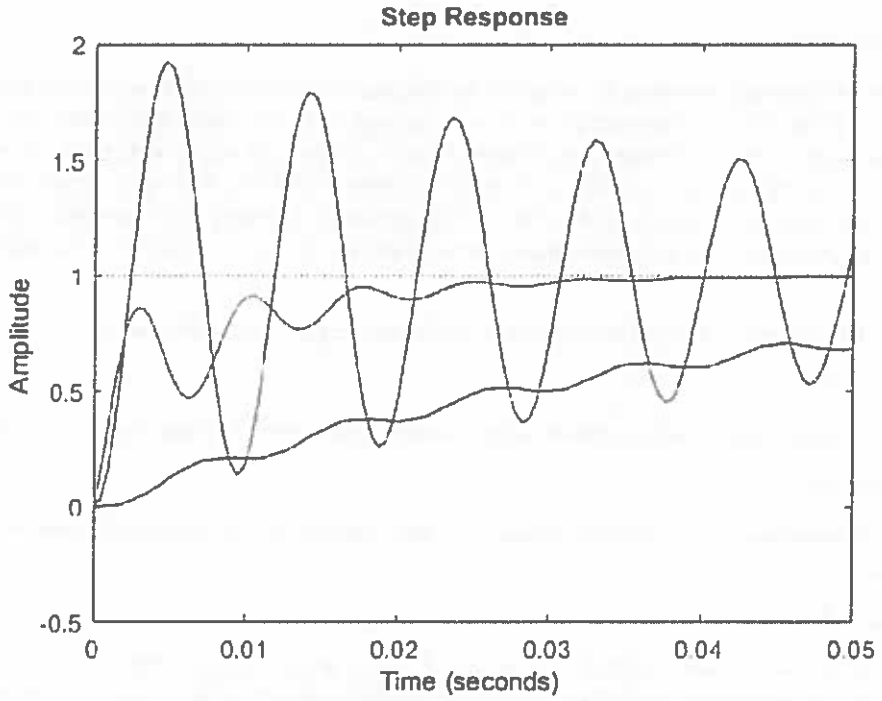


Figure 5: Red line is the open loop step response normalised by its steady state value. Blue line is the closed loop step response using a PID controller. Black line is the closed loop step response using a PI controller.

As the step plot above shows, using a PID controller it is possible to obtain a much faster response whilst still limiting the percentage overshoot to 0. The settling time now is approx. about 0.028s i.e. now a factor of 10 times shorter than using only a PI controller in closed loop or the step response of the open loop system.

7.3 – Closed Loop PSpice Model

We opted to use op-amps because we couldn't get the Laplace model block on Pspice to function properly. We kept getting extremely high voltages (e.g. a few MVI). Five op-amps would be needed in total to realise our compensator i.e. a difference amplifier to produce the error signal, an integrator to provide zero steady state error, a differentiator to control the damping factor of the

system i.e. to provide a good trade-off between percentage overshoot (< 10%) and speed of response, a gain stage to provide a signal proportional to the error and a summing amplifier to add all these terms together. Our compensator designed using matlab unfortunately didn't work as expected on Pspice. The factors multiplying each error component were far too small.

Schematic of closed loop converter circuit with matlab designed compensator implemented:

See appendix C

Traces showing matlab designed compensator step response at start-up for full load operation (i.e. 80Ω):

See appendix D

Traces showing matlab designed compensator step response at start-up for half load operation (i.e. 160Ω):

See appendix D

For the compensators, we designed using Matlab, the step responses obtained for both full and half load on start-up had some percentage overshoot of approx. 8.3%, there were no oscillations as expected compared with the open loop step response. In addition, the steady state error was non-zero i.e. the output voltage was approx. 38V instead of 48V. The settling time was in excess of 14ms for full load operation and twice as long for half load operation. Overall, the compensator designed in Matlab produced a very poor performance. It only satisfied one of the requirements i.e. <10% overshoot.

Schematic of closed loop converter circuit with improvised compensator implemented:

See appendix C

Traces showing improvised compensator step response at start-up for full load operation (i.e. 8Ω):

See appendix D

Traces showing improvised designed compensator step response at start-up for half load operation (i.e. 16Ω):

See appendix D

By trial and error, we could construct a much better compensator on Pspice which met all the requirements for both full and half load operation. This compensator had a zero-steady state error (i.e. Vset = 48V = Vout), zero percentage overshoot, a much faster speed of response i.e. a rise time of about 3ms, and it also provided some limiting to the input current in rush by introducing damping into the system. With this compensator, we could keep the max. input current in rush below 15A.

Implemented compensator transfer function:

$$H(s) = \frac{0.0024s^2 + 10s + 5 * 10^5}{s(2 * 10^{-5}s + 1)}$$

Poles are at 0 and approx. -50000
Zeroes are at approx. $-2.08 * 10^3 \pm j1.43 * 10^4$

This compensator has a completely different transfer function from the compensator obtained using the matlab PID design tool but it has the same necessary elements to meet the given closed loop

design requirements i.e. an integrator and a phase lead component. The proportional term had little effect on shaping the step response hence it was omitted. The differentiator was made to be slightly lossy by introducing a real pole far away to the left-hand side from the jω axis. This was necessary to produce a smooth output signal (instead of abrupt delta lines) to avoid Pspice producing the error, "failed to converge".

Proportional term of improvised compensator = 0

Derivative term of improvised compensator $= \frac{0.0024s}{2*10^{-5}s+1)}$

Integral term of improvised compensator = $\frac{5*10^3}{s}$

Appendix:

Appendix A: Various Half-Bridge Drive Circuit topologies:

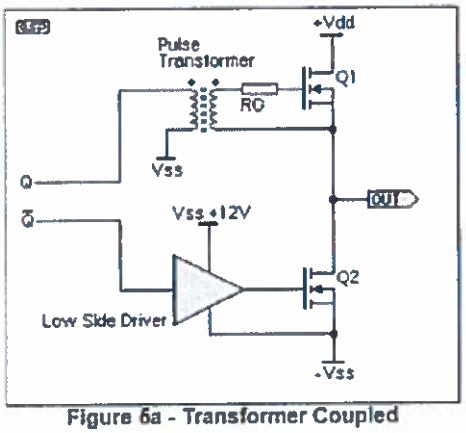


Figure 6a - Transformer Coupled

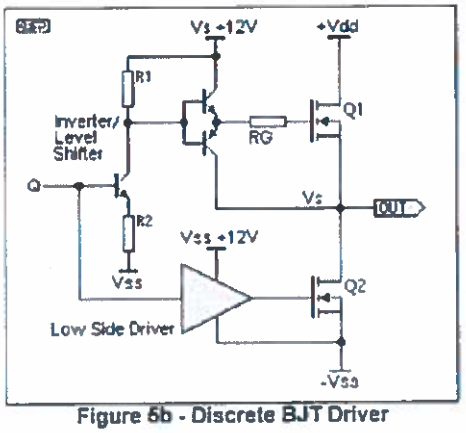


Figure 6b - Discrete BJT Driver

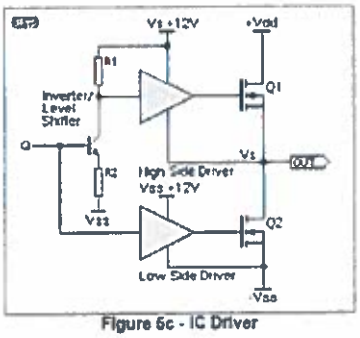


Figure 6c - IC Driver

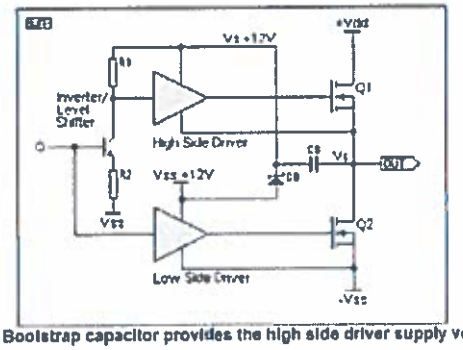


Fig. 6 Bootstrap capacitor provides the high side driver supply voltage

Appendix B: State Space Averaging model:

$$\dot{\tilde{x}} = A\tilde{x} + E\tilde{w}$$
$$\tilde{y} = C\tilde{x} + F\tilde{w}$$

$$\delta = \Delta + \tilde{\delta}$$
$$x = X + \tilde{x}$$
$$y = Y + \tilde{y}$$
$$u = U + \tilde{u}$$

Where $\tilde{w} = [\tilde{\delta}]$, $\tilde{x} = \begin{bmatrix} \tilde{I_L} \\ \tilde{V_C} \end{bmatrix}^T$, and $\tilde{y} = [V_o]$. The tilde accent represents a perturbation of the variable at a specific operating point. δ, I_L, V_C, and V_o have their usual meanings i.e. duty cycle, inductor current, capacitor voltage and output voltage respectively.

$$A = \Delta A_{on} + (1 - \Delta)A_{off}$$
$$B = \Delta B_{on} + (1 - \Delta)B_{off}$$
$$C = \Delta C_{on} + (1 - \Delta)C_{off}$$
$$D = \Delta D_{on} + (1 - \Delta)D_{off}$$
$$E = (A_{on} - A_{off})X + (B_{on} - B_{off})U$$
$$F = (C_{on} - C_{off})X + (D_{on} - D_{off})U$$
$$X = A^{-1}BU$$

The MOSFET, inductor and capacitor are taken to be ideal hence the inductor and capacitor series resistance are set to 0Ω. Hence:

$$A_{on} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \qquad B_{on} = B_{off} = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \qquad C_{on} = C_{off} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \qquad A_{off} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix}$$
$$D_{on} = D_{off} = [0]$$

Where R represents the resistance of the load, L represents the inductance of the inductor and C represents the capacitance of the inductor.

Using the matrices above the full small signal state space model system matrices can be formulated based on a set duty cycle operating point (Δ) of 7/12.

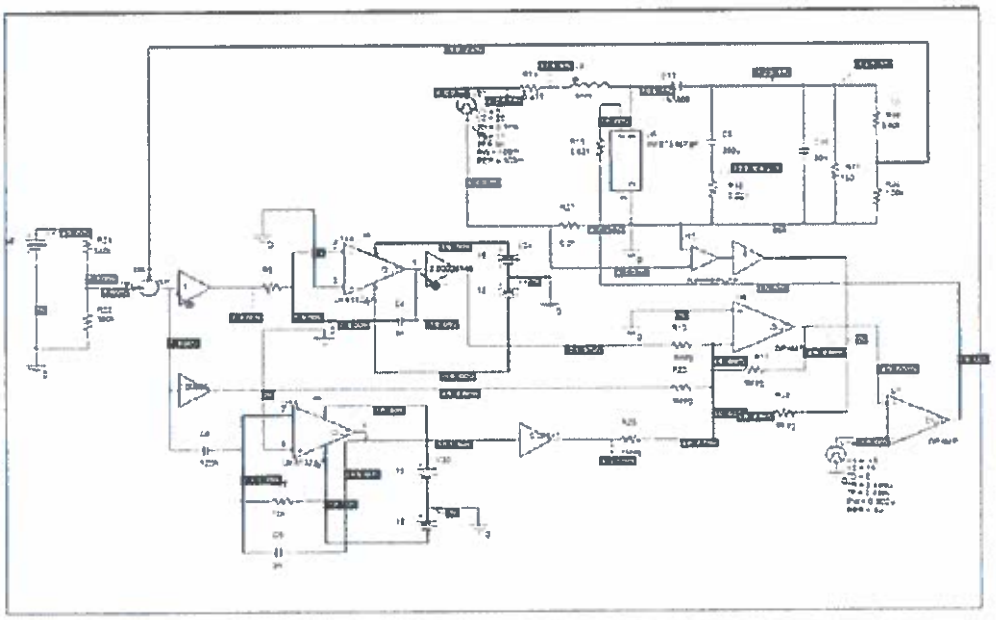
$$A = \begin{bmatrix} 0 & -\frac{1-\Delta}{L} \\ \frac{1-\Delta}{C} & -\frac{1}{RC} \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \qquad C = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \qquad D = [0]$$

Using the component values we chose (i.e. L = 1mH and C = 390μF), the state space matrices were:

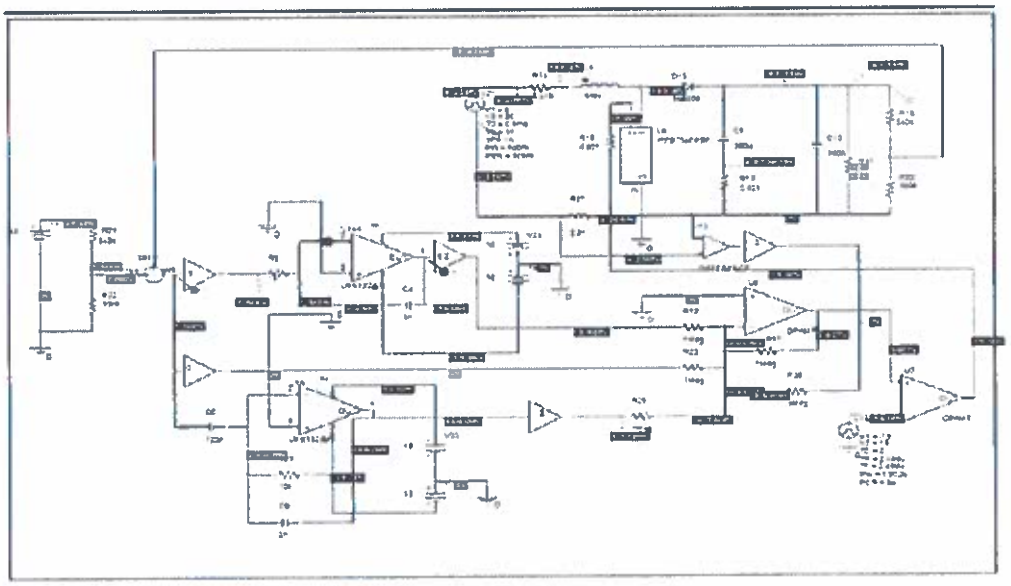
$$A = \begin{bmatrix} 0 & -416.7 \\ 1068 & -32.05 \end{bmatrix} \qquad E = \begin{bmatrix} 4.8 * 10^4 \\ -3692 \end{bmatrix} \qquad C = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \qquad F = [0]$$

Appendix C: Schematic for closed loop converter circuit with different compensator designs:

Schematic of closed loop converter circuit with matlab designed compensator implemented:



Schematic of closed loop converter circuit with improvised compensator implemented:



Appendix D: Pspice traces for compensator designs:

Trace descriptions:

Green trace is Vout

Yellow trace is the output from the differentiator

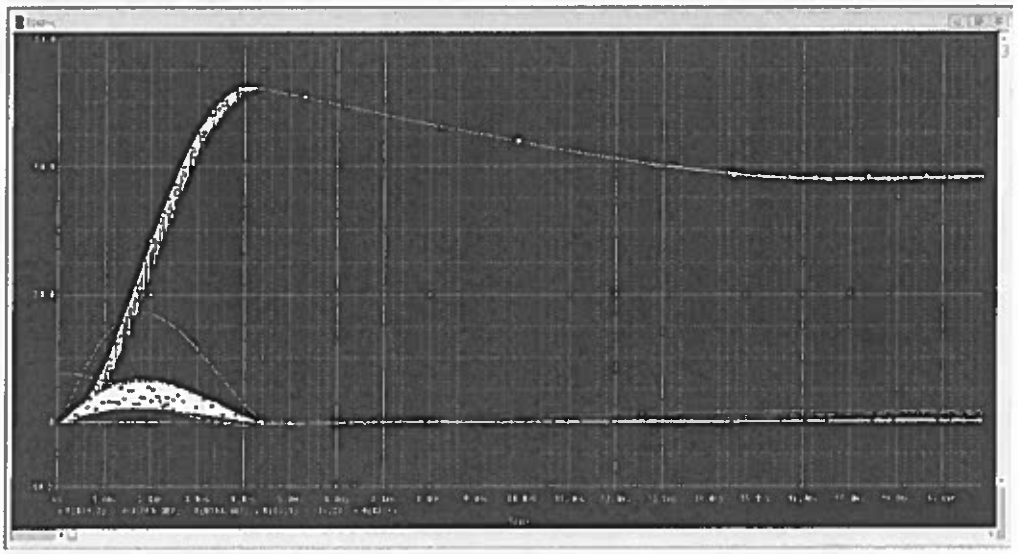
Blue trace is the output from the integrator

Pink trace is the input current or inductor current

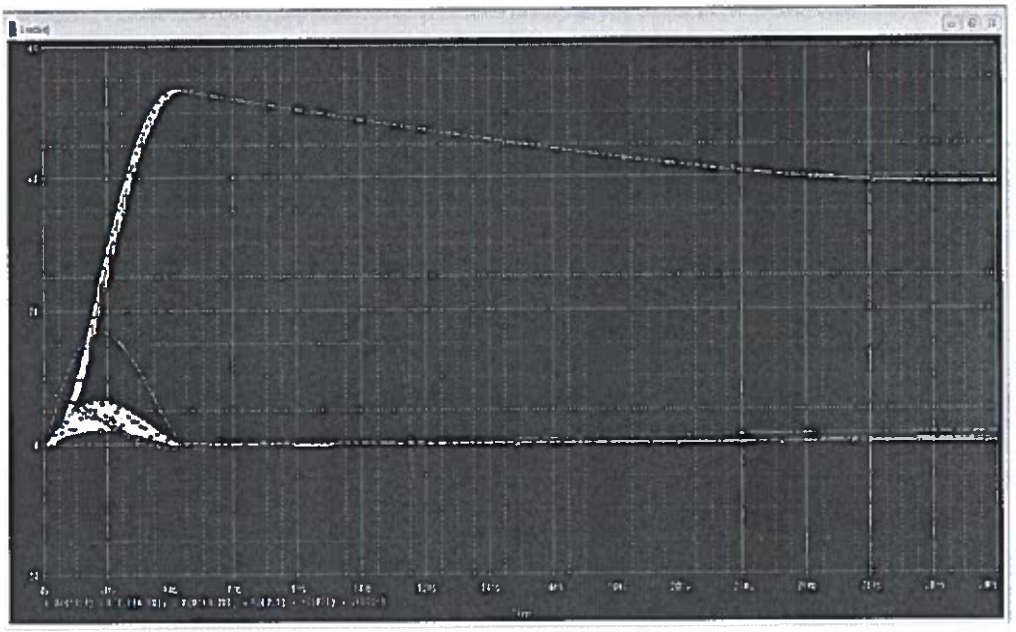
Red trace is the error signal

Cyan trace is the modulating signal used to generate the PWM output to drive the MOSFET

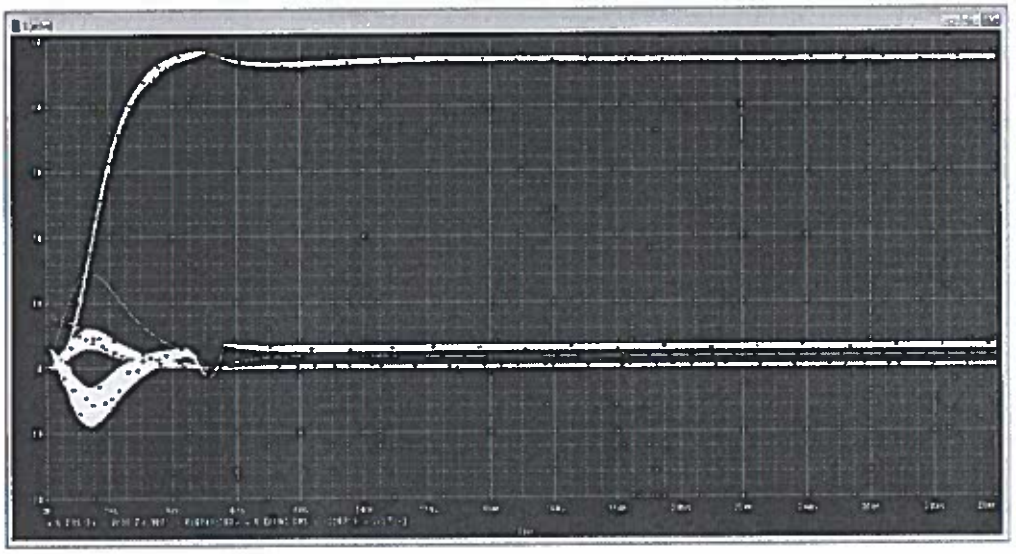
Traces showing matlab designed compensator step response at start-up for full load operation (i.e. 80Ω):



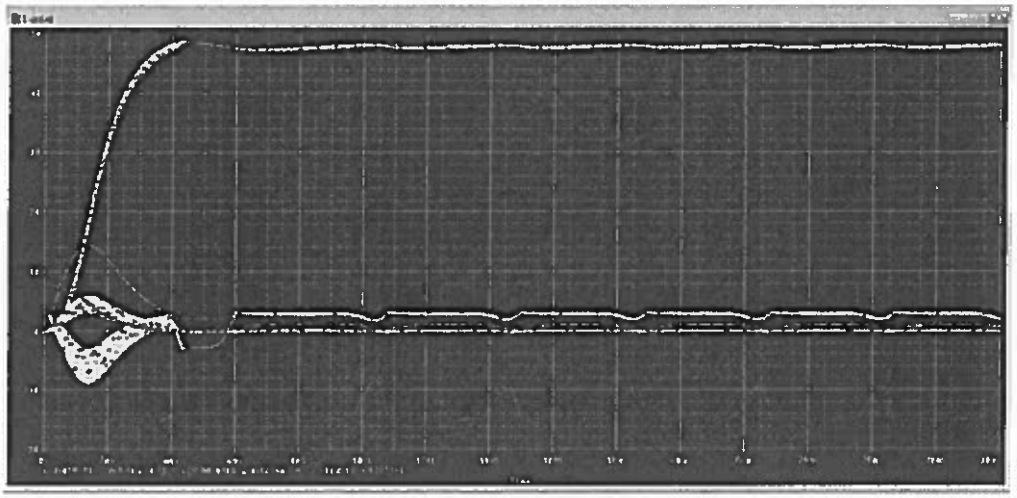
Traces showing matlab designed compensator step response at start-up for half load operation (i.e. 160Ω):



Traces showing improvised compensator step response at start-up for full load operation (i.e. 80Ω):



Traces showing improvised designed compensator step response at start-up for half load operation (i.e. 160Ω):



Appendix E:

Table 1 : Device Efficiency with Different Loads with Ideal Components

R load (Ohm)	80	160	240	320	640	1280
Vout steady (V)	43.337	44.2	44.501	44.6555	44.895	45.0255
Iin steady (mA)	1246.4	634.037	425.2065	320.888	159.42	80.7325
Iout steady (mA)	540.033	276.248	185.42	139.548	70.149	35.176
Efficiency	93.88%	96.29%	97.03%	97.10%	98.77%	98.09%

Table 2: Detailed measurement with R load max and steeped load change response with Ideal Components

For Rout max = 80 Ohm				
	Reading 1	Reading 2	Average	Delta
Vout steady (V)	43.339	43.335	43.337	0.004
One cycle time(ms)	23.84	23.845		0.005
Oscillation Frequency(kHz)			200	
I(R1) (mA)	270.87	270.846	270.858	0.024
I(R2) (mA)	269.187	269.163	269.175	0.024
I(total out) (mA)	540.057	540.009	540.033	0.048
V max (V)	47.753	47.749	47.751	
Rise time (ms)	4.9835	4.9852	4.98435	
Settling time (ms)	12.023	12.025	12.024	
Vout Percentage Overshoot			10.18%	
I(in) max (A)	9.704	9.675	9.6895	
I (in) steady (A)	1.2725	1.2203	1.2464	0.0522
efficiency			93.88%	
Stepped response from 100% to 50%				
Vout steady (V)	44.199	44.201	44.2	0.002
I(R1) (mA)	276.242	276.254	276.248	0.012
I(R2) (uA)	44.192	44.194	44.193	0.002
I(in) steady (mA)	661.165	606.909	634.037	54.256
efficiency			96.29%	

Table 3: Detailed measurement with R small with Ideal Components

	Reading 1	Reading 2	Average	Delta
Vout steady (V)	38.827	38.814	38.8205	0.013
Oscillation Freqency(kHz)			200	
I(out) (A)	1.9407	1.9414	1.94105	0.0007
V max (V)	42.202	42.187	42.1945	
Rise time (ms)	4.9835	4.9852	4.98435	
Settling time (ms)	10.398	10.402	10.4	
Vout Percentage Overshoot			8.69%	
I(in) max (A)	10.7	10.674	10.687	
I (in) steady (A)	4.4766	4.4342	4.4554	0.0424
efficiency			84.56%	

Table 4: Detailed measurement with R load max and steeped load change response with Real Components

For Rout max = 80 Ohm				
	Reading 1	Reading 2	Average	Delta
Vout steady (V)	42.5	42.527	42.5135	0.027
Vout spike (V)	42.345	42.533		0.188
V max (V)	54.298	54.341	54.3195	
Rise time (ms)	6.6819	6.6839	6.6829	
Settling time	17.738	17.742	17.74	
Vout Percentage Overshoot			27.77%	
I(R17) (mA)	263.995	264.202	264.0985	0.207
I(R2) (mA)	265.645	265.853	265.749	0.208
I(total out) (mA)	529.64	530.055	529.8475	0.415
I (out) spike (mA)				1.15
I(in) max (A)	17.926	17.955	17.9405	
I(in) min (mA)				-4.9385
t for I(in) reach 0 (ms)				
I (in) steady (A)	1.2284	1.2607	1.24455	0.0323
One cycle time(ms)	24.95	24.955		0.005
Oscillation Freqency(kHz)				
efficiency			90.50%	
I diode spike (A)	5.7	-6.85		12.55
Stepped response from 100% to 50%				
Vout steady (V)	43.217	43.2	43.2085	0.017
Vout spike (V)	43.217	43.12		0.097
I(R1) (mA)	270.113	270	270.0565	0.113
I(R2) (uA)	43.211	43.193	43.202	0.018
I(in) steady (mA)	668.898	616.505	642.7015	52.393
I (out) spike (mA)				0.098
efficiency			90.78%	
I diode spike (A)	3	-4.12		7.12

Table 5: Device Efficiency with Different Loads with Real Components

R load (Ohm)	80	160	320	640	1280
Vout steady (V)	42.5155	43.2085	44.1125	45.365	47.287
Iout steady (mA)	529.8475	270.0565	137.85	70.883	36.9425
Iin steady (mA)	1244.55	642.7015	335.5475	176.75	94.2085
Efficiency	90.50%	90.78%	90.61%	90.96%	92.71%

Table 6: Detailed measurement with R small with Real Components

	Reading 1	Reading 2	Average	Delta
Vout steady (V)	38.737	38.611	38.674	0.126
Vout spike (V)	38.737	38.2		0.537
V max (V)	47.878	47.721	47.7995	
Rise time (ms)	6.6819	6.6839	6.6829	
Settling time (ms)			16.021	
Vout Percentage Overshoot			23.60%	
I(Rout) (A)	1.9343	1.9406	1.93745	0.0063
I (out) spike (mA)			25.8	
I(in) max (A)	18.255	18.226	18.2405	
I(in) min (A)			1.3809	
t for I(in) reach min (ms)			8.4652	
I (in) steady (A)	4.5799	4.5327	4.5563	0.0472
One cycle time(ms)	24.95	24.955		0.005
Oscillation Freqency(kHz)			200	
efficiency			82.23%	
I diode spike (A)	4.5626	-18.336		22.8986

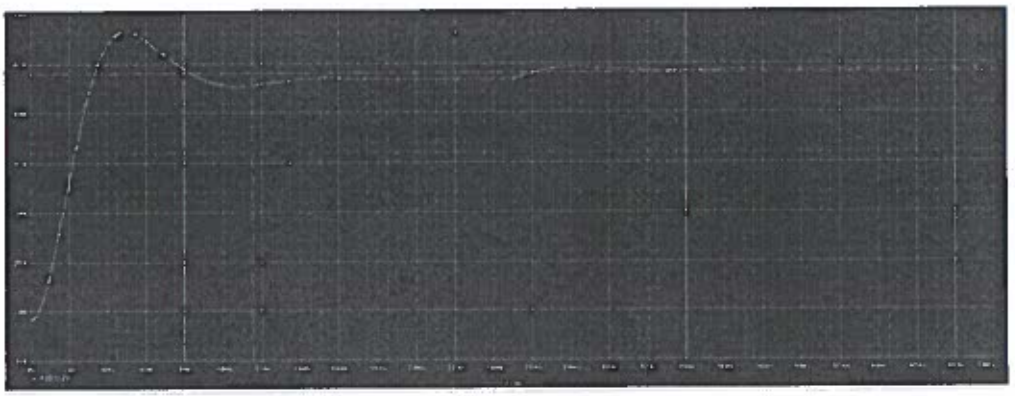
Table 7: Device Efficiency with Different Loads with Real Components and Synchronous Rectification

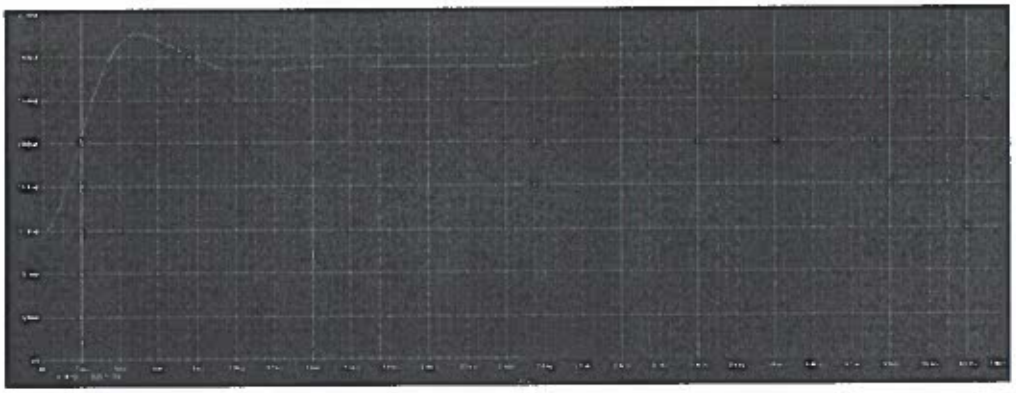
R load (Ohm)	80	160	320
Vout steady (V)	42.992	43.894	44.879
Iout steady (mA)	1260	661.3705	349.887
Iin steady (mA)	537.386	274.3395	142.26
Efficiency	91.04%	91.04%	91.24%

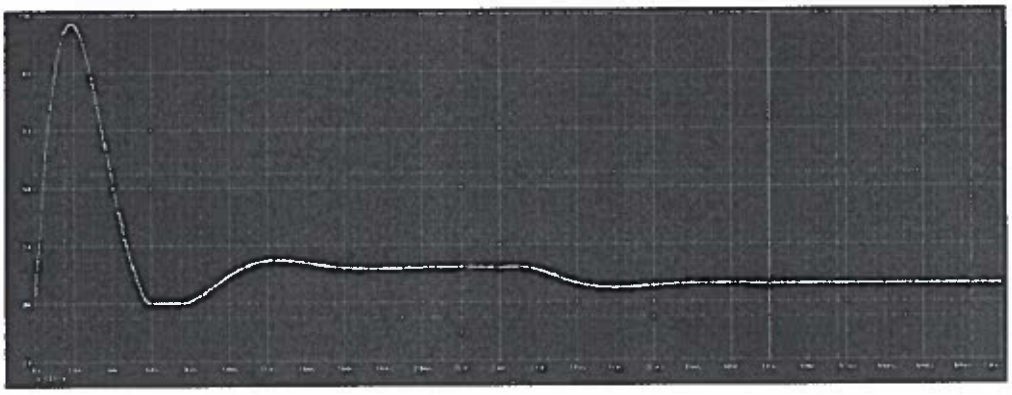
Table 8: Detailed measurement with R load max with Real Components and Synchronous Rectification

	Reading 1	Reading 2	Average	Delta
Vout steady (V)	42.975	43.009	42.992	0.034
Vout spike (V)	42.78	43.011		0.231
V max (V)	54.889	54.936	54.9125	
Rise time (ms)	54.889	54.936	54.9125	
Settling time (ms)			14.072	
Percentage Overshoot			27.73%	
I(Rout) (mA)	537.166	537.606	537.386	0.44
I (out) spike (mA)	537.592	534.743		2.849
I(in) max (A)	18.245	18.216	18.2305	
I(in) min (mA)	-588.683	-528.083	-558.383	
t for I(in) reach 0 (ms)			7.0398	
t for I(in) reach min (ms)	7.79	7.793	7.7915	
I (in) steady (A)	1.2861	1.2339	1.26	0.0522
One cycle time(ms)	24.95	24.955		0.005
Oscillation Freqency(kHz)			200	
efficiency			91.68%	
I diode-MOSFET steady state(A)	1.2594	2.15E-06		1.259398
I diode-MOSFET (A)	3.7245	-11.473		15.1975
V diode-MOSFET min (mV)	-684.042			
V diode-MOSFET drop time (ns)	-39.773			
V diode-MOSFET rise time (ns)	-283.144			

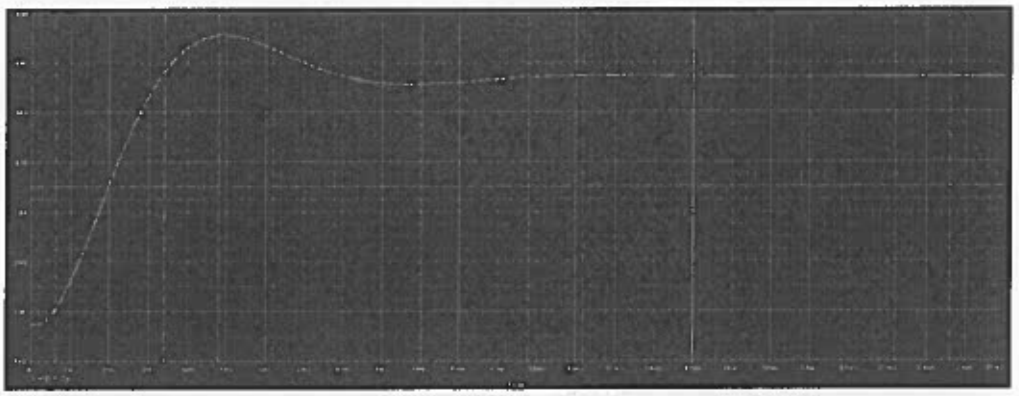
Appendix F1: Pspice Simulation for R load max and steeped load change response with Ideal Components

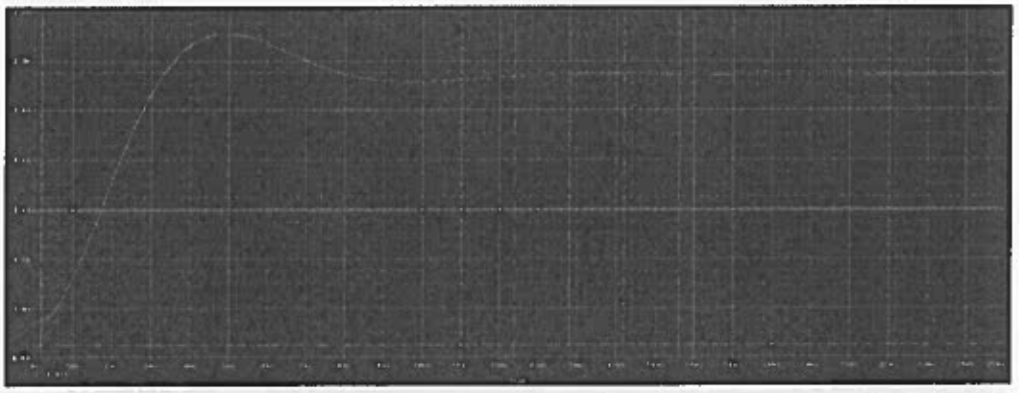


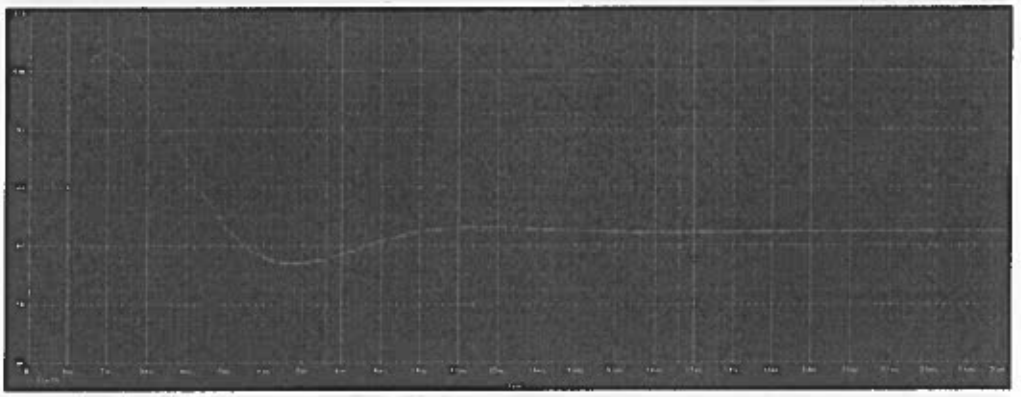




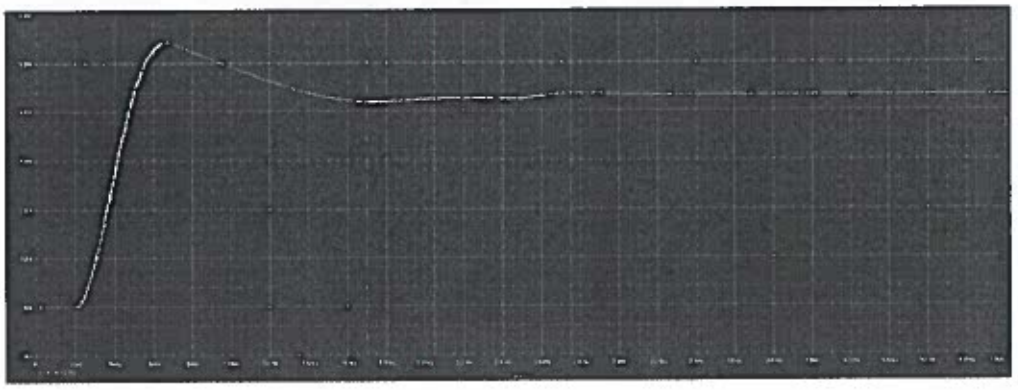
Appendix F2: Pspice Simulation for Low R load with Ideal Components

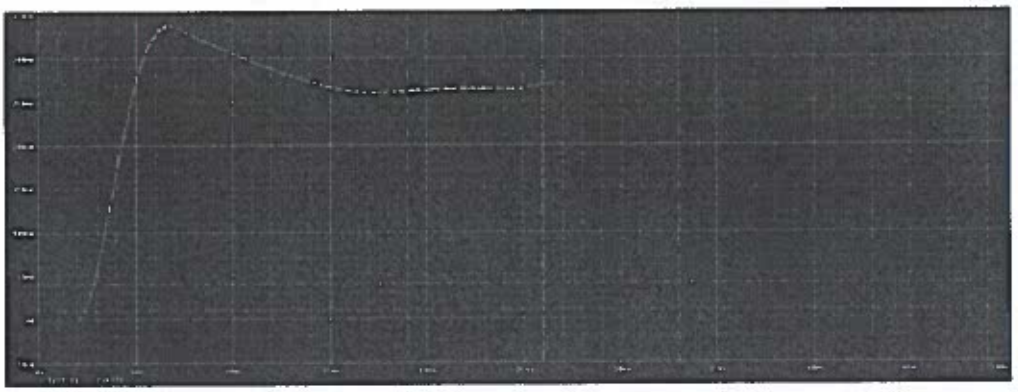


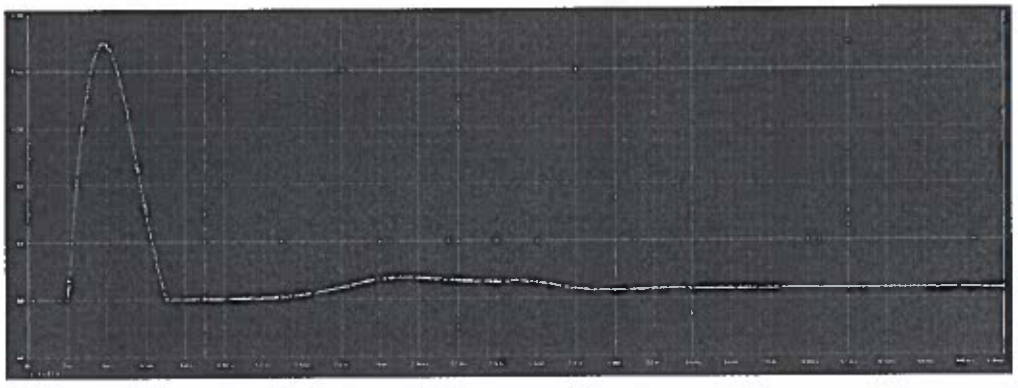




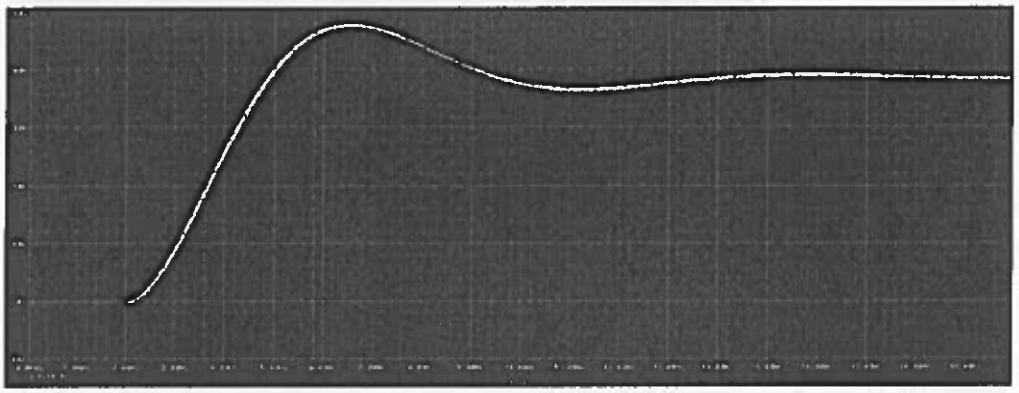
Appendix F3: Pspice Simulation for R load max and steeped load change response with Real Components

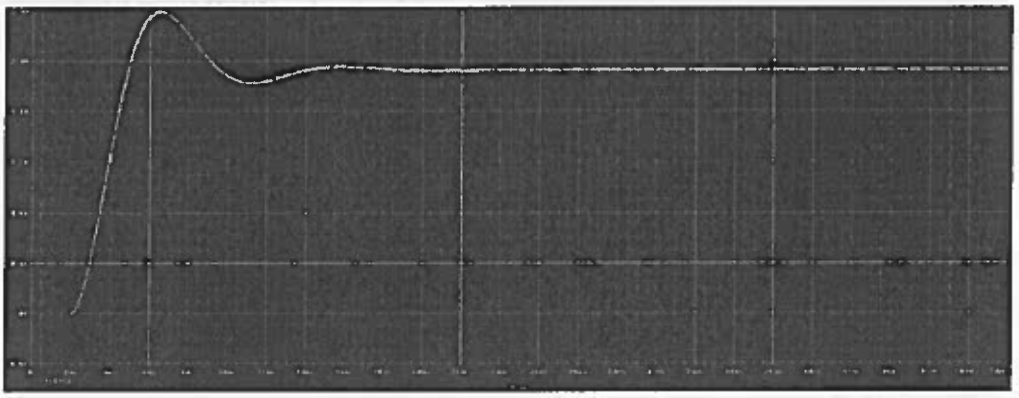


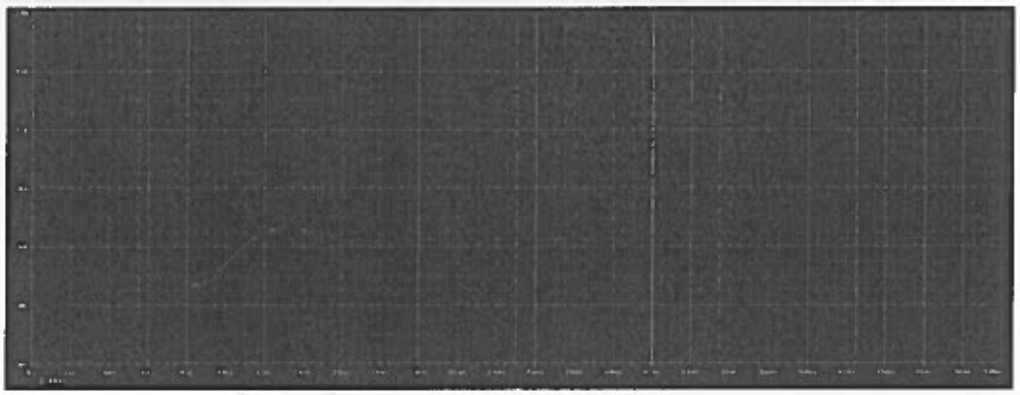




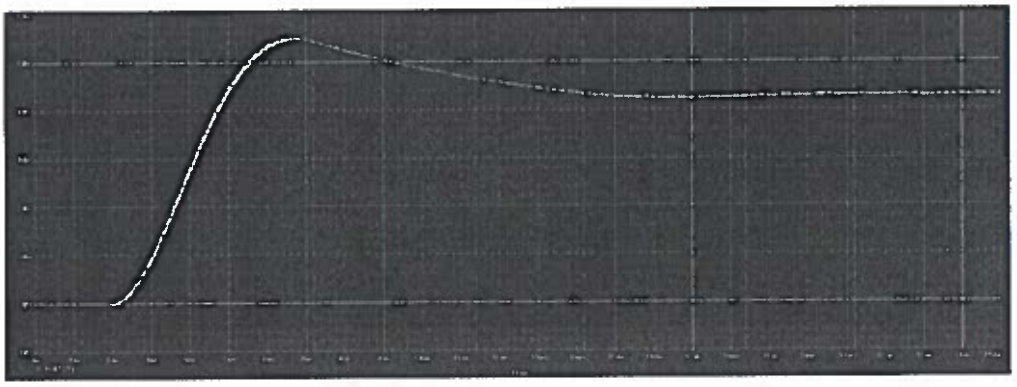
Appendix F4: Pspice Simulation for Low R load with Real Components

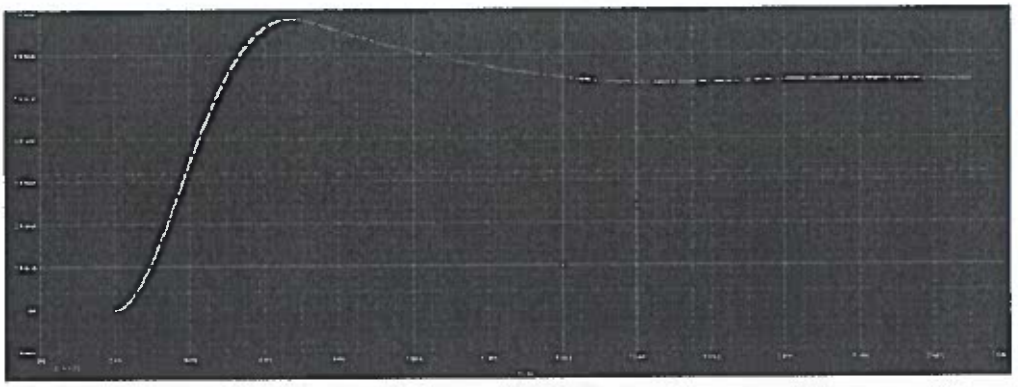


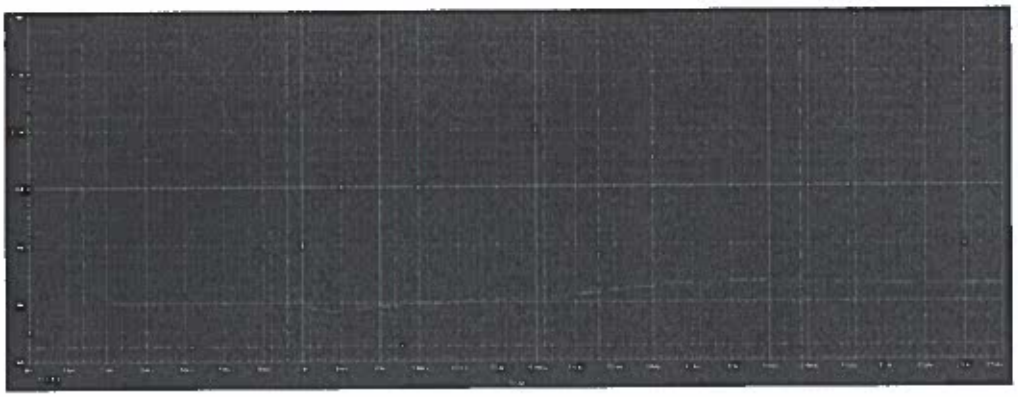




Appendix F5: Pspice Simulation for Max R load with Real Components and Sych. Rectifyer







Power Electronics Report2

GRADEMARK REPORT

FINAL GRADE

/100

GENERAL COMMENTS

Instructor

PAGE 1

Comment 1

Total grade: C

Converter design: A
Open Loop: E
Component Choices: C
Real Components: A
Synchronous Rectification: B
Closed Loop: D
Additional Marks: E

PAGE 2

Comment 2

Why?

Comment 3

Good analysis

Comment 4

Good

Comment 5

It is better to double these specs in hard switching topologies (as this one) and work with 100V ratings

PAGE 3

Comment 6

If C was calculated at 1.5uF and L at 365uH, why are these values not in the simulation?

Comment 7

This is not what the appendix is for

Comment 8

Again. Information left in the appendix will only be marked if further proof of a method of calculation or something similar if necesary

PAGE 4

Comment 9
Only if you analyse steady state

Comment 10
What is low?

Comment 11
!

Comment 12
good!

Comment 13
Not always

Comment 14
Nope!

PAGE 5

Comment 15
Depends on the transistor and how you drive it.

Comment 16
Important in appendix?

PAGE 6

Comment 17
?

PAGE 7

Comment 18
and failure

Comment 19
Did you consider the losses of the driver?

PAGE 8

PAGE 9

Comment 20
There is a problem on doing this. You should find an expression that depends on the

variables of the system to identify what varies when a perturbation (dIo or dVdc) is present in the system

PAGE 10

QM

Citation Needed

Cite Source:
Please use the link below to find links to information regarding specific citation styles:
http://www.plagiarism.org/plag_article_citation_styles.html

Comment 21

This has nothing to do with the system model

Comment 22

You can include it!

QM

Vague

Unclear:
When making a point in one of your body paragraphs, one of the most common mistakes is to not offer enough details. A paragraph without much detail will seem vague and sketchy. A paper is always strengthened when your claims are as specific as possible, The more detailed evidence you offer, the more reference points your reader will have. Remember that you are communicating your argument to a reader who has only your description to go by. Someone who reads your essay will not automatically know what you mean to express, so you have to supply details, to show the reader what you mean, not just tell him or her.

Comment 23

eliminate

Comment 24

If you increase the phase margin you will just make the system more robust, it has nothing to do with steady state error

PAGE 11

Comment 25

This is true, but the system will become slower

PAGE 12

Comment 26

This should not be in the report

PAGE 13

Comment 27

This can be optimised to 2

Comment 28

It's common for these things to require a little bit of tuning in the end but you should always know what you are designing

PAGE 14

Comment 29

If two different tools present heavily different results, it means that an error was present when designing this. It is not enough to say that there is an error. It should be found

PAGE 15

PAGE 16

PAGE 17

PAGE 18

PAGE 19

PAGE 20

PAGE 21

PAGE 22

PAGE 23

PAGE 24

PAGE 25

PAGE 26

PAGE 27

PAGE 28

PAGE 29